

Promoting Early Science Learning with Dialogic AI

Summary of Proposed Work

This request for supplement funding is submitted in response to the Dear Colleague Letter (DCL) contained within solicitation NSF 25-035.

For the last six years, with three NSF AISL grants (Nos. 1923136, 2115382, 2415882), we have been developing educational media incorporating dialogic uses of artificial intelligence to promote young children's science learning. Our studies indicate that children who use these dialogic media to engage in AI-powered conversation about the content significantly improve in their science learning, language development, and school readiness compared to children who access non-dialogic versions of the same media (see, e.g., Xu, Aubele, et al., 2022; Xu, Vigil, et al., 2022; Xu et al., 2024). In our most recent AISL grant (2415882), we have been working with parents in the Santa Ana Early Learning Initiative to develop a series of science e-books suitable for children ages 4-5. These interactive books incorporate AI-based dialogic questions for children designed to reinforce learning of STEM concepts and practices, as well as family discussion prompts to spark further dialogue with adults. Twelve books for this series will be completed this fall and twelve more will be completed by next summer for a total of 24.

Though these AI-powered e-books have been developed for out-of-school use, they represent an "out-of-the box" resource that can be used with even broader impact in public schools. The science content of these books is being designed in accord with the California Preschool/Transitional Kindergarten Foundations under the leadership of co-PI Andres Bustamante, who served as the lead content expert for the California Department of Education committee that developed the new early science learning standards. This proposal thus calls for a two-year supplemental research and development project, utilizing design-based implementation research, to further develop, pilot, evaluate, and scale up these materials for use in classrooms.

For this project, we will build on our ongoing work to address the following aims:

Aim 1. Iteratively pilot, evaluate, and refine the platform for use in TK-K classrooms to promote children's learning of early science through an AI-powered dialogic reading platform.

Aim 2. Create professional development and curricular materials to support teaching of early science and AI literacy through an AI-powered dialogic reading platform.

Aim 3. Scale the platform for national dissemination to improve science knowledge and AI literacy.

Project Timeline

The 24-month timeline of this study will be as follows:

1. Initial Pilot (Phase 1): January 1 to July 31, 2026
 - a. Pilot the 12 e-books in Grades TK-K in El Sol Science and Arts Academy, a partner school in our Elementary Computing Research Practice Partnership (RPP)
 - b. Working with the El Sol teachers, develop curricular and professional development materials for effective integration, including materials to support AI literacy development of both students and teachers
 - c. Develop a teacher dashboard for the e-book platform so that instructors can get better information on how children are using it
 - d. Integrate the additional 12 e-books into the platform
2. Expanded Pilot (Phase 2): August 1, 2026 to June 30, 2027
 - a. Pilot the 24 e-books with teacher dashboard in Grades TK-K in El Sol and six other schools in the three partner school districts of the Elementary Computing RPP (two schools each in

- Santa Ana Unified, Fullerton School District, and Montebello Unified), beginning with professional development for the teachers there
- b. Evaluate children's learning processes and outcomes, and continue to refine the curriculum and professional development materials to best support children's learning
 - c. Port the PD materials to an online system for nationwide access
 - d. Start recruiting for broad state and national distribution
3. Scale-up (Phase 3): July 1, 2027 to December 31, 2027
- a. Make platform nationally available
 - b. Online professional development also nationally available
 - c. Monitor and support national usage

Below, we expand on the significance of the proposed project, specifications of the platform and its content, and our plans for iterative refinement, evaluation, scaling, and dissemination of the platform for broad classroom use.

Leveraging AI-Powered Dialogic Reading for Improved Science Learning

At the core of our intervention is *AI-powered dialogic reading*. Dialogic reading is an interactive form of shared reading in which an adult engages the child in back-and-forth conversation during reading—asking questions related to story content and providing contingent feedback based on the child's responses (Whitehurst & Lonigan, 1998). Decades of research has demonstrated that this interactive approach to reading with young children results in improved language and literacy outcomes when compared to traditional non-interactive reading (Mol et al., 2008; Mol et al., 2009; Towson et al., 2017; What Works Clearinghouse, n.d.). Our research has demonstrated that AI-powered conversational agents can carry out this approach, effectively engaging children in conversation during reading—and that conversational agents can confer comparable benefits to that of a human reader applying the same dialogic approach (Xu, Aubele, et al., 2022). Furthermore, our research has demonstrated that children who engage with AI-enhanced interactive science videos show better science learning outcomes than counterparts watching non-interactive versions (Xu, Vigil et al., 2022; Xu, He et al., 2024). By providing children with personalized feedback and incorporating scaffolding, conversational agents carry the potential to improve children's learning at scale in a way that can powerfully supplement teachers' instructional practices.

The current project extends this promising line of research through the integration of AI-powered science e-books into early childhood classrooms to enhance children's science learning and promote AI literacy. Under our current AISL grant (2415882), we have been working with local parents to co-design science e-books that are compelling and developmentally appropriate for young children. The e-books, twelve of which will be complete this fall, feature story content and embedded *conversational moments* that are grounded in the California Preschool/Transitional Kindergarten Learning Foundations (PTKLF) for Science. These learning standards, developed by a national team of experts in early childhood science education (led by co-PI Bustamante), closely align with the Next Generation Science Standards (Next Generation Science Standards [NGSS], n.d.a.), and represent the cutting edge in foundational scientific knowledge and skills for young learners. The PTKLF covers five key strands: 1) science and engineering practices; 2) physical science; 3) life science; 4) earth and space science; and 5) engineering, technology, and applications of science.

Through rich narratives, our e-books integrate cross-cutting concepts across science strands for comprehensive coverage. As one example, an ebook currently under development centers on a young boy growing corn in his apartment. As the boy tests different approaches to facilitating its growth, he models scientific and engineering practices (PTKLF Strand 1). For instance, as the protagonist experiments with varying the amount of water given to the corn plant and records observations of the consequent impact on plant characteristics, students learn about practices such as making observations (1.1), asking questions (1.3), planning and carrying out investigations (1.6), and using tools (1.7).

Simultaneously, students learn foundational life science concepts (Strand 3), such as the needs of living things (3.7) and their life cycles (3.6).

The conversational moments, embedded intermittently throughout the book, reinforce these concepts through repetition of key ideas, scaffolding, and fostering of children's engagement. For instance, in one conversational moment to be integrated into the corn story, the AI agent asks the student to reflect on what happened when the boy did not give enough water to the plant, and to propose a hypothesis about what will happen if the boy gives more water. In this way, the conversational agent reinforces the aforementioned scientific concepts, and also teaches the meanings of general academic and science vocabulary words (e.g., *observation*, *hypothesis*, *growth*). In sum, the platform is rooted in evidence-based practices, capitalizing on limited classroom time by integrating science and literacy (e.g., Hwang et al., 2022), while also providing young learners a chance to use AI responsibly for learning in a structured environment.

Building on the Platform for Classroom Use: Teacher Dashboard & Curricular Materials

Under our current AISL grant (2415882), we have developed a platform for at-home use to enable AI-powered dialogic reading. As noted, the platform will be ready for immediate deployment, loaded with 12 complete science e-books this fall, and an additional 12 e-books by the end of July 2026 (i.e., before the beginning of Phase 2). In addressing Aim 1 of the project, a key focus will be iterative development of the platform and curricular materials to support classroom use.

The e-book platform constitutes an “out-of-the-box” tool that can seamlessly be integrated into classrooms as part of their “station learning” routines—a common practice whereby students rotate in small groups to different stations for learning activities. In its current version optimized for at-home use, the platform features science e-books with conversational moments, during which the conversational agent asks questions related to science content and vocabulary and provides contingent feedback. Through automated scaffolding mechanisms integrated into the conversational agent design, and togglable features (e.g., audio narration of book text), the platform can facilitate implementation across classroom contexts and provide teachers and students with flexibility to ensure the platform meets their needs.

To further support classroom use, however, it is important that teachers have access to data to monitor students' platform usage, and other relevant progress metrics. Hence, during Phase 1, we will develop a teacher dashboard for the platform. This will equip teachers with real-time student engagement analytics, including basic usage metrics (e.g., time spent on the platform and number of books read), along with AI conversation quality metrics to provide detailed insight into student verbal engagement with the AI agent during reading sessions. Moreover, the dashboard will feature classroom management tools for book assignment and progress monitoring to enable teachers to implement the platform as part of a structured curricular approach. Additional features may be developed, tested, and refined during the initial and expanded pilot phases based on stakeholder input (i.e., feedback from teachers at El Sol).

To enhance support for learning of science concepts covered in the e-books, we will also collaborate with teachers to develop curricular materials. For instance, currently, the e-books include a “science activities” component, where children and their families are encouraged to try their own experiments or projects at home (e.g., growing their own plant at home). Some of these activities will be adapted for classroom implementation to afford students opportunities to apply their learning in real-world contexts. Finally, we will create curricular resources to support young children's learning about AI.

Professional Development

During Phase 1, we will co-create professional development materials with El Sol teachers through monthly co-design sessions. The PD will focus on three areas: 1) navigating the platform and implementing it in the classroom, along with corresponding curricular materials; 2) developing knowledge of the California Preschool/Transitional Kindergarten Learning Foundations for Science, and how to

effectively apply these standards to guide teaching practices; and 3) AI literacy. As AI has become ubiquitous in students' daily lives and will play an increasingly important role in their future academic and professional development, there is an urgent need to provide early childhood educators with AI literacy skills, and effective strategies for implementing AI tools in the classroom. Indeed, there is currently a notable lack of age-appropriate AI literacy resources in primary education (Su et al., 2023). Hence, to strengthen the impact of our platform and address the goals of the Executive Order on Advancing Artificial Intelligence Education for American Youth, we will develop an AI literacy component for our PD. Our lab already has extensive experience developing AI literacy materials for K-12 contexts.

Project Design and Evaluation

To assess project success, we will conduct an evaluation during Phase 2 (August 2026 to June 2027), with deployment of all 24 e-books across seven schools: El Sol Science and Arts Academy plus six additional schools from our Elementary Computing RPP (two each from partner districts). We will employ a mixed-methods design combining quantitative outcome measures with rich process data automatically collected through the platform and qualitative insights from observations and teacher interviews. Successful implementation of the platform will be evaluated, guided by the following questions:

1. To what extent is the AI-powered dialogic e-book platform implemented in classrooms with fidelity?
2. To what extent does the platform impact science learning outcomes in TK-K classrooms?
3. To what extent do the professional development materials effectively prepare teachers?

Through the platform, we will automatically gather real-time usage data, which will give insight into reading frequency and patterns in book usage throughout the school year. We will also collect conversation log data to analyze students' verbal engagement with the conversational agent during their reading sessions. We will then code these data for features associated with learning. Notably, our previous research has demonstrated that verbal engagement (e.g., response rate, relevance, and accuracy) during conversational moments is significantly correlated with science learning outcomes (Xu et al., 2024).

At the end of each ebook, we will also use the conversational agent to embed short assessments of science knowledge and science vocabulary words. These built-in learning indicators will align with the PTKLF. By automating collection of these data through the platform, we will be able to efficiently collect short-term learning outcome data at scale.

Finally, we will conduct classroom observations during implementation and administer surveys to participating teachers both pre- and post-implementation to gain insight into their experiences with the platform, integrating AI tools into their classroom instruction in general, and the effectiveness of our professional development approach. Qualitative observational and interview data will be thematically analyzed (Braun & Clarke, 2019) to identify key factors supporting successful AI tool integration and areas for improvement in the professional development materials.

Together, these metrics will provide deep insight into the educational impact of our platform and the associated materials. Informed by these data, we will refine the platform for Phase 3 (Scale-up; July 2027 to December 2027) for broad national dissemination. During this time, we will make the platform content and professional development materials freely available online. We will also focus efforts on disseminating our findings widely to different stakeholders (e.g., families, researchers, policymakers, and teachers) through our website, social media channels, traditional academic research venues (e.g., conferences), and our partnership networks to spread awareness of best practices on integrating AI into primary classrooms and improving young children's science learning.

Leveraging Partnership Networks for Efficient and Sustainable Implementation

For this project, we will rely on our well-established network of research-practice partners. This will ensure that the platform can be adopted in classrooms for prolonged use in a way that meets the needs of the teachers and their students. During Phase 1, our main partner for this project will be El Sol Santa Ana Science and Arts Academy, a charter school—located near the UCI School of Education—with whom we have a long-standing partnership and have collaborated on multiple large-scale research projects. The school comprises approximately 50 students in transitional kindergarten and 100 students in kindergarten (California Department of Education, 2025).

During the expanded pilot phase, we will work with two additional schools each from our three partner districts (two schools each in Santa Ana Unified, Fullerton School District, and Montebello Unified) in our Elementary Computing RPP. Our well-established relationships with these school districts will enable us to efficiently carry out Phase 2, and will provide insight into platform implementation and impact across varying school contexts.

Finally, toward the end of Phase 2, in preparation for Phase 3, we will recruit broadly through our numerous partners. Specifically, we will seek to scale the platform nationally and disseminate materials and findings through our Elementary Computing Research Practice Partnership, Orange County Educational Advancement Network, Orange County Department of Education AI Committee, AI for California, and Common Sense Media. This will enable us to carry out an ambitious scale-up phase in a time- and cost-effective manner.

Intellectual Merit

Though AI-based dialogic media have proven effective in promoting children's science and language development in out-of-school settings, they have not yet been tested in school classrooms. Our research findings will contribute essential knowledge to the field regarding best practices for integrating AI-enhanced media and AI literacy instruction into young children's curriculum, addressing an urgent question in educational research and informing policy decisions about AI implementation in primary education.

Broader Impact

Early years are critical for students developing the science knowledge and AI literacy skills valuable for future success. However, currently there are insufficient STEM materials suitable for grades TK and K. For example, in California, universal transitional kindergarten was just recently introduced, and there are very few curricular materials available for science, and virtually none for AI literacy.

This project will generate significant educational and research outcomes with national implications for AI integration in early elementary classrooms. Through systematic implementation and evaluation across school contexts, we anticipate AI literacy development and measurable improvements in science knowledge and vocabulary. Beyond immediate student outcomes, our professional development materials, science e-book platform, and curricular materials will be made freely available, providing educators across the nation with evidence-based resources for incorporating AI-enhanced instruction into their classrooms.

Justification for Supplemental Funding

The early years are critical for providing students a strong foundation in science knowledge and AI literacy, but there are few curricular materials available for that. This request leverages substantial prior NSF investments in creating out-of-school materials for improving students' science learning and AI literacy (i.e., NSF Grants Nos. 1923136, 2115382, 2415882), to now adapt them for in-school use. In particular, the 24 AI-powered science e-books we are creating now under NSF Grant No. 2415882, for which we request this supplement, represent an "out-of-the-box" resource that can be used in TK-K grades without need for sustained external personnel, engineers, programmers, or other technical support staff. The budget request will support adapting this resource for public education through small one-time improvements (e.g., creating a teacher dashboard and some curricular activities); piloting the resource in schools; and developing online PD materials so that teachers nationwide can access and use the e-books.

The budget items are all designed to support this adaptation. The major item will be for personnel, including a graduate student researcher with expertise in early childhood education to carry out the pilot and help develop professional development materials, a small amount for a software engineer to develop the dashboard, and modest levels of additional support for the two PIs to successfully manage this additional work. Other funds requested are for small stipends for participating teachers and travel costs to local schools.

Finally, the supplemental project will address all four themes discussed in the Dear Colleague Letter on Expanding K-12 Resources For AI Education

1. **Teacher professional development:** We will carry out teacher professional development for effective use of these AI-enhanced STEM resources in grades TK-K.
2. **Curricula and instructional materials:** We will also develop the appropriate curricula and instructional materials to facilitate this effective use.
3. **Technology and tools:** We will begin with a technology platform previously developed with NSF funding that is ready for "out-of-the-box" use, and then refine, evaluate, and scale it for the broadest possible impact in TK-K classrooms.
4. **Networks:** We will leverage multiple extant networks to promote the project, including our Elementary Computing Research Practice Partnership group of four local educational agencies, the Orange County Educational Advancement Network, the Orange County Department of Education AI network, and the AI for California statewide organization.